Group Forecast and Furious
Casey Barrasso, Joshua Previte, Geovanni Jones

TARGET VARIABLE

We are trying to predict the directional movement of AAPL stock between periods. Therefore, our target variable will be a binary indicator that represents the directional change in closing price between consecutive trading periods. These will be defined as 1 for an increase and 0 for a decrease or no change. The directional change will be calculated using returns from the previous period. In our case, this will be returns from the previous trading day. Returns are used instead of raw prices because they use a stationary representation that better captures the market's reaction to information such as new sentiment. The returns variable will be constructed using the raw time-series closing price data, and the directional change binary indicator will be constructed using the returns variable. This allows us to build classification models that use news sentiment to predict the directional change of the stock price between periods.

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PREDICTORS

The primary predictor variable in this study is the news sentiment score derived from articles related to AAPL. For each time period, news articles are aggregated to produce a sentiment polarity score ranging from −0.99 to 1.00, where negative values indicate negative sentiment, positive values indicate positive sentiment, and values near zero indicate neutral sentiment. Because the objective of this study is to examine the impact of new information on stock returns, representing sentiment as a continuous numerical variable provides a principled and interpretable way to incorporate news sentiment into the predictive models.

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EXPLORATION OF THE DATASET

Apple Stock (AAPL): Historical Financial News Data obtained from kaggle.com is the news data which we will be working with for the purpose of this project. This dataset contains financial news data that is related to Apple Inc. (AAPL) and includes information from multiple news articles and their corresponding metadata, sentiment analysis, and relevance to Apple stock.

Dataset Description:

- Date: The timestamp of the news article in ISO 8601 format (international standard for writing dates and times so that everyone, humans and computers, understands them the same way, no matter the country (data type is a datetime object)
- Title: The headline of the news article (data type is a string).
- Content: The body text of the article (data type is a string).
- Link: The URL link to the full news article (data type is a string).
- Symbols: Stock symbols related to the content (e.g., AAPL, MSFT) (data type is a string).
- Tags: Additional keywords or categories assigned to the article (data type is a string).

Sentiment Analysis:

- Sentiment Polarity: An overall measure of sentiment in the article, ranging from negative to positive (data type is a float).
- Sentiment Negative: The proportion of negative sentiment in the text (data type is a float).
- Sentiment Neutral: The proportion of neutral sentiment in the text (data type is a float).
- Sentiment Positive: The proportion of positive sentiment in the text (data type is a float).

Key Features:

- Rows: 29,752 entries, each representing a unique news article.
- Columns: 10 features providing metadata, content, and sentiment analysis.
- Sentiment Coverage: Sentiment values are available for 29,737 entries.

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The Apple Stock dataset contains historical daily share prices of Apple Inc. (AAPL) over a long time period. Each observation represents the closing share price of Apple stock for a specific trading day.

Variables in the dataset:

- Dates: The trading date for each observation (daily frequency) (data type is a datetime object).
- AAPL Share Price: The closing price of Apple's stock on that date, measured in U.S. dollars (data type is a float).

The dataset includes 5,298 daily observations, indicating that it spans many years of trading history, from Apple's early growth phase to its position as a mature, high-value company. Because the data is recorded daily, it is well suited for time-series analysis, trend analysis, and volatility studies.

Overall Price Level

- Minimum price: $1.13
- Maximum price: $286.19

Central Tendency

- Mean (average) price: $62.29
- Median (50th percentile): $27.06

With a mean value that is much higher than the median value, this would suggest that Apple's stock price was relatively low for a large portion of the dataset and very high prices which were seen in later years pulled the average upwards. This also indicates a right-skewed distribution which is common in successful stock growths.

Volatility

- Standard deviation: $73.45

The large standard deviation indicates high variability in Apple's stock price over time. This is expected as the dataset covers many years and Apple is a company that experienced rapid innovation, market expansion, stock splits, and major product releases over the course of the years recorded in the dataset.

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REFERENCE

- https://www.kaggle.com/datasets/frankossai/apple-stock-aapl-historical-financial-news-data